

Skills Summary

Comparing Unlike Fractions: Benchmarks and Common Denominators

Use this reference while completing your worksheet or when studying.

Two fractions can only be compared when they count the *same size pieces of the same whole*. That is the one idea behind all four skills below: a benchmark compares each fraction to a piece you already know, and a common denominator makes the pieces literally the same size.

1. Compare with a benchmark (benchmark-comparison)

Rule: A fraction is **less than** $\frac{1}{2}$ when the numerator is less than half the denominator, **equal** when it is exactly half, and **greater** when it is more.

So: half of the denominator is the number to beat. If one fraction lands below $\frac{1}{2}$ and the other above it, you are done — no common denominator needed.

Example: Compare $\frac{3}{7}$ and $\frac{5}{8}$.

Step 1. The two fractions look close, so test each one against $\frac{1}{2}$ before doing any arithmetic with denominators.

Step 2. Half of 7 is 3.5, and $3 < 3.5$, so $\frac{3}{7} < \frac{1}{2}$. Half of 8 is 4, and $5 > 4$, so $\frac{5}{8} > \frac{1}{2}$. Below the benchmark is smaller than above it: $\frac{3}{7} < \frac{5}{8}$

Try it: Compare $\frac{2}{9}$ and $\frac{4}{7}$ using the benchmark $\frac{1}{2}$.

check: $\frac{4}{4} > \frac{6}{4}$

2. Compare with a common denominator (common-denominator)

Rule: Rewrite both fractions with the same denominator — any common multiple works, the least one keeps the numbers small — then compare the numerators.

$\frac{a}{b} = \frac{a \times k}{b \times k}$ Same denominator means same size pieces, so more pieces is more.

Example: Compare $\frac{3}{4}$ and $\frac{7}{10}$.

Step 1. Both fractions are above $\frac{1}{2}$, so the benchmark cannot separate them — rebuild them on a common denominator.

Step 2. 20 is a multiple of both 4 and 10: $\frac{3}{4} = \frac{15}{20}$ and $\frac{7}{10} = \frac{14}{20}$, and $15 > 14$. $\frac{3}{4} > \frac{7}{10}$

Try it: Compare $\frac{5}{6}$ and $\frac{8}{9}$ on a common denominator.

check: $\frac{9}{8} > \frac{9}{8}$

3. Build an equivalent fraction (equivalent-fractions)

Rule: Multiply the numerator *and* the denominator by the same number. That is multiplying by $\frac{k}{k} = 1$, so the amount does not change — only the size and count of the pieces do. To reach a target denominator, ask “*times what?*” and use that same factor on top.

Example: Write $\frac{2}{3}$ with denominator 12.

Step 1. The target denominator is given, so find the factor that turns 3 into 12 and apply it to both parts.

Step 2. $3 \times 4 = 12$, so multiply the numerator by 4 as well: $\frac{2 \times 4}{3 \times 4} = \frac{8}{12}$

Try it: Write $\frac{3}{5}$ with denominator 30.

check: $\frac{18}{30}$

4. Order three or more fractions (ordering-sets)

Rule: Put *every* fraction on one common denominator, order the numerators, then write the answer back in the original fractions — the question asked about those, not about the rewritten ones.

Example: Order $\frac{2}{5}$, $\frac{1}{3}$, $\frac{1}{2}$ from least to greatest.

Step 1. Three fractions cannot be compared in pairs reliably, so convert all three to one denominator at once.

Step 2. 30 works for 5, 3, and 2: $\frac{2}{5} = \frac{12}{30}$, $\frac{1}{3} = \frac{10}{30}$, $\frac{1}{2} = \frac{15}{30}$; the numerators order $10 < 12 < 15$.
 $\frac{1}{3} < \frac{2}{5} < \frac{1}{2}$

Try it: Order $\frac{3}{4}$, $\frac{5}{6}$, $\frac{2}{3}$ from greatest to least.

check: $\frac{9}{12} > \frac{8}{12} > \frac{5}{12}$

(!) Watch out: A bigger denominator does *not* mean a bigger fraction — it means smaller pieces. With the same numerator, $\frac{1}{10} < \frac{1}{8}$. And never compare by numerators alone: $\frac{5}{12}$ is less than $\frac{2}{3}$ even though $5 > 2$.
